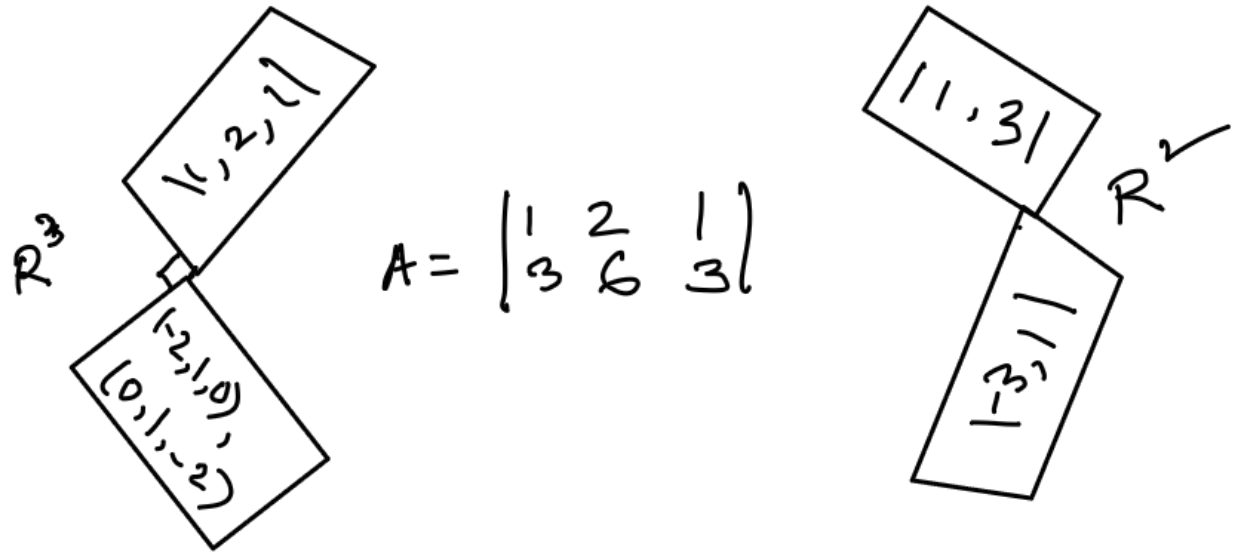
Ch 4.1

① 2×3 Mat of rank 1. ∴ $r = 1$, $m = 2$, $n = 3$

$n - r = 3 - 1 = 2$ (nullspace's dimension)

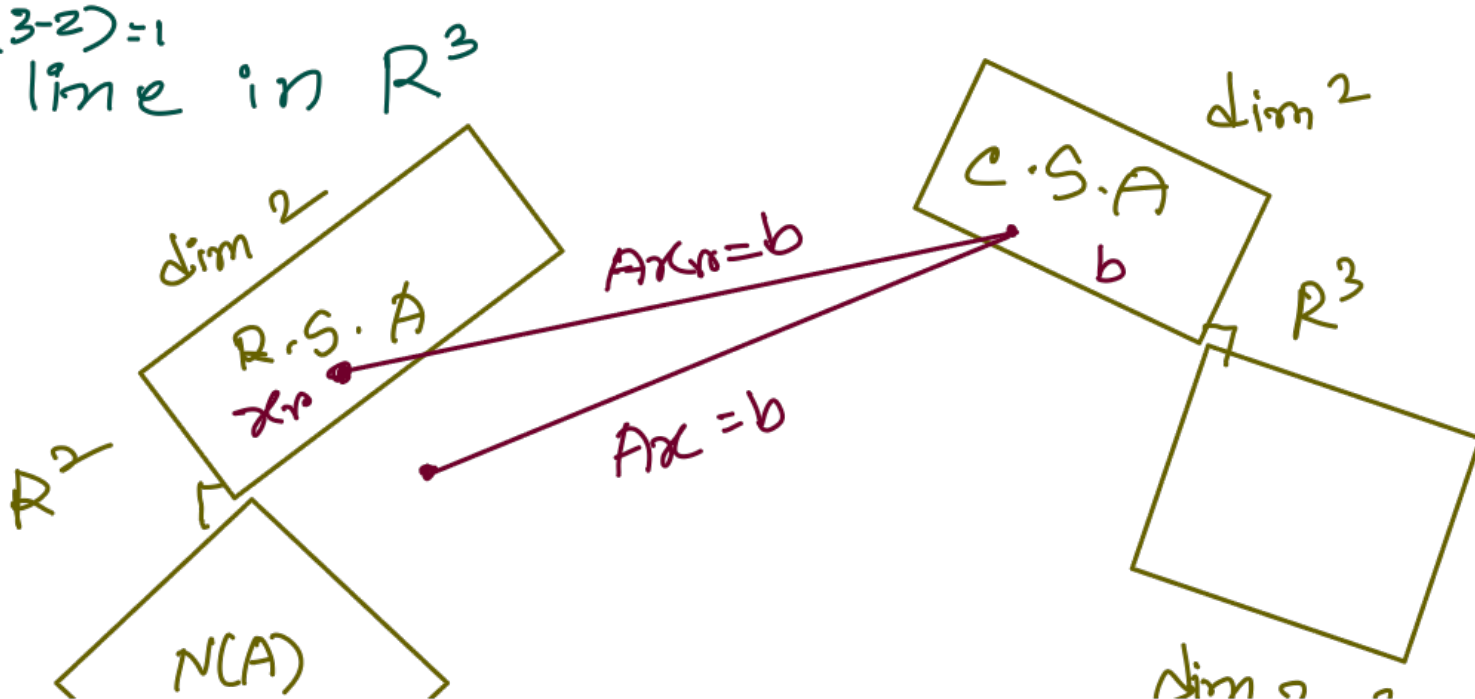
2 nullspace vectors, orthogonal to rowspace vectors in R^3 (R^n). colspace & nullspace of A^T are perpendicular lines in R^2.



2 3×2 Mat of rank 2.

colspace = plane perpendicular to left nullspace

$(3-2) = 1$ = line in R^3



dim 2-2

a) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & -3 & -1 \\ -3 & 5 & -2 \end{bmatrix}$ ✓

b) $\begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix}$ is not orthogonal to $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ ✗

c) $Ax = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ has soln and $A^T \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$\Rightarrow$ $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ in colspace of A.

$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ in left nullspace of A.

$Ax = 0$ → in left nullspace of A

$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ must be orthogonal.

check → $1 \cdot 1 + 0 + 0 = 1 \neq 0$ fails. ✗

d) $\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$ $\begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1 - 1 = 0$

every row orthogonal to every col. ✓

e) cols add up to col of 0's.

rows " rows " 1's

For 3x3,

To get sum of col multiply by $\begin{bmatrix}1\\1\\1\end{bmatrix}$

$$\underset{\text{(Mat A)}}{[\ ]}\begin{bmatrix}1\\1\\1\end{bmatrix}_{1\times 3} = \quad .$$

So $\begin{bmatrix}1\\1\\1\end{bmatrix}$ is in nullspace of A.

and $[1\ 1\ 1]$ is in rowspace.

$$\begin{bmatrix}1\\1\\1\end{bmatrix}[1\ 1\ 1] = 1+1+1 = 3 \neq 0$$

R(A) & N(A) orthogonality fails.

④ If $AB=0$, then cols of B are in the nullspace of A. $(Ax=0)$

rows of A's in left nullspace of B.

if A is a 3x3 Mat, the Null space dimension

$$N(A) = 3-2 = 1$$

B lies in nullspace of A. So all 3 col of B must be linear combination of a single vector. So B's rank at most is 1.

5(a) $Ax = b$ (has sol^n) , $A^Ty = 0$ So y is $\perp$ to b

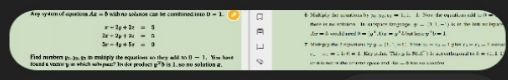
b: is in colspace of A ; $A^Ty = 0$: in left nullspace

$b^Ty = 0 \Rightarrow (Ax)^Ty = 0$; $x^TA^Ty = 0$

$y^Tb = 0$ ✓ $y^TAx = 0$

(b) $A^Ty = (1,1,1)$ has a sol^n $\longrightarrow$ $(1,1,1)$ is a combination of the rows of A. So in rowspace. $Ax = 0$ $\hookrightarrow$ in nullspace of A.

$\therefore y^Tx = 0$



explain 6

The problem discusses a system of linear equations:

$$
\begin{aligned}
x + 2y + 2z &= 5 \\
2x + 2y + 3z &= 5 \\
3x + 4y + 5z &= 9
\end{aligned}
$$

The key idea is that the system has no solution, meaning the right-hand side vector b is not in the column space of A. To show inconsistency, we need to find a set of multipliers y_1, y_2, y_3 such that the left-hand side sums to zero but the right-hand side sums to a nonzero value (0 = 1 contradiction).

By choosing $y_1 = 1$, $y_2 = 1$, and $y_3 = -1$, we multiply the equations:

1. $(1) \cdot (x + 2y + 2z = 5) \quad \rightarrow x + 2y + 2z = 5$
2. $(1) \cdot (2x + 2y + 3z = 5) \quad \rightarrow 2x + 2y + 3z = 5$
3. $(-1) \cdot (3x + 4y + 5z = 9) \quad \rightarrow -3x - 4y - 5z = -9$

Adding these,

$$(x + 2y + 2z) + (2x + 2y + 3z) + (-3x - 4y - 5z) = 5 + 5 - 9$$

$$0 = 1$$

This contradiction shows that the system has no solution.

Subspace Interpretation

The vector y = $(1, 1, -1)$ is in the left null space of A, meaning it satisfies $y^T A = 0$. However, since $y^T b = 1 \neq 0$, it indicates that b is not in the column space of A, proving inconsistency.

(8) $x = x_r + x_n$ (x_n: in nullspace)

$Ax_n = 0 \therefore Ax = Ax_r$

Only rowspace components lead to result.

$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ $x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Project x onto rowspace:

=> Find x_r in the form $x_r = k\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Solve for k by minimizing the distance bet^n x & rowspace by using projection formula:

$$k = \frac{x^T v}{v^T v} = \frac{\begin{bmatrix} 1 & 0 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix}}{\begin{bmatrix} 1 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix}} = \frac{1}{1+1} = \frac{1}{2}$$

$$\therefore x_r = \frac{1}{2}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

$$x_n = x - x_r = \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$$

$$= \begin{bmatrix} 1/2 \\ -1/2 \end{bmatrix}$$

(12) $x - x_r = x_n$; $(2, 0) - (1, -1) = (1, +1)$

(13) Every vector v in V orthogonal to every vect w in W



no planes can be orthogonal in R^3

14 $Ax = B\hat{x} \quad \therefore \quad Ax - B\hat{x} = 0$

$$[A \; B] \begin{bmatrix} x \\ -\hat{x} \end{bmatrix} = 0$$

3 homo eqns, 4 unknowns } always a nonzero soln.

$$\begin{bmatrix} 1 & 2 & 5 & 4 \\ 1 & 3 & 6 & 3 \\ 1 & 2 & 5 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = 0$$

each col space is a plane in R^3 → nontrivial soln

$$\begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 1 & 2 \end{bmatrix}$$

$n \rightarrow R^n = R^3$

$3 \times 2 \rightarrow$ plane

colspace

The intersection of the 2 planes is at least a line. → nonzero vect.

$v = Ax = B\hat{x} \neq 0$

12:49 Fri, 4 Apr 74%

Nullspace and Subspace Analysis Explained

To solve problem 14, we need to find a vector common to both column spaces of matrices A and B. Here's the process:

1. Set up the equation:
   We seek vectors $\mathbf{x} = (x_1, x_2)$ and $\tilde{\mathbf{x}} = (\tilde{x}_1, \tilde{x}_2)$ such that:

$$A\mathbf{x} = B\tilde{\mathbf{x}}.$$

   This translates to solving:

$$\begin{cases} x_1 + 2x_2 = 5\tilde{x}_1 + 4\tilde{x}_2 \\ x_1 + 3x_2 = 6\tilde{x}_1 + 3\tilde{x}_2 \\ x_1 + 2x_2 = 5\tilde{x}_1 + \tilde{x}_2 \end{cases}$$

2. Solve the system:

The equations are consistent and underdetermined (3 equations, 4 unknowns). By inspection or substitution, one solution is:

$$\mathbf{x} = (3, 1), \quad \tilde{\mathbf{x}} = (1, 0).$$

Substituting into $A\mathbf{x}$ and $B\tilde{\mathbf{x}}$:

$$A\mathbf{x} = 3\begin{bmatrix}1\\1\\1\end{bmatrix} + 1\begin{bmatrix}2\\3\\2\end{bmatrix} = \begin{bmatrix}5\\6\\5\end{bmatrix}, \quad B\tilde{\mathbf{x}} = 1\begin{bmatrix}5\\6\\5\end{bmatrix} = \begin{bmatrix}5\\6\\5\end{bmatrix}.$$

3. Geometric interpretation:
   - The column spaces of A and B are 2-dimensional subspaces (planes) in $\mathbb{R}^3$.
   - By the dimension formula, if $\dim(V) + \dim(W) > n$, their intersection has dimension $\geq \dim(V) + \dim(W) - n$. Here, $2 + 2 > 3$, so their intersection is at least a line.
   - The vector $(5, 6, 5)$ spans this line of intersection.
4. Non-orthogonality:
   Since the planes share a line, they are not orthogonal subspaces (orthogonal subspaces intersect only at the origin).

Final Answer

The vector $\begin{bmatrix}5\\6\\5\end{bmatrix}$ is in both column spaces, confirming their intersection is a line.

Thus, the solution is:

$$\boxed{\begin{bmatrix}5\\6\\5\end{bmatrix}}$$

New chat

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DeepThink (R1) Search

12:51 Fri, 4 Apr 74%

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1. The Freedom of Choice in an Underdetermined System

In the equation

$$A\mathbf{x} = B\tilde{\mathbf{x}},$$

This means there is not a unique solution—we have a family of solutions.

A common strategy in such cases is to choose one of the vectors arbitrarily (as long as it is not the zero vector), then solve for the other vector.

---

2. Choosing $\tilde{\mathbf{x}} = (1, 0)$

The choice $(1, 0)$ for $\tilde{\mathbf{x}}$ is made because it is simple and nonzero. This choice helps simplify the equations:

- It sets $\tilde{x}_1 = 1$ and $\tilde{x}_2 = 0$, reducing the number of unknowns in the system.
- With these values, the equations become directly solvable for $\mathbf{x}$.

---

3. Substituting $\tilde{\mathbf{x}} = (1, 0)$ into the Equations

The original system (obtained from equating $A\mathbf{x}$ and $B\tilde{\mathbf{x}}$) was:

LaTeX Copy code

```
\begin{cases}
x_1 + 2x_2 = 5\tilde{x}_1 + 4\tilde{x}_2,\quad1mm]
x_1 + 3x_2 = 6\tilde{x}_1 + 3\tilde{x}_2,\quad1mm]
x_1 + 2x_2 = 5\tilde{x}_1 + \tilde{x}_2.
\end{cases}
```

Substitute $\tilde{x}_1 = 1$ and $\tilde{x}_2 = 0$:

- First Equation:
  $x_1 + 2x_2 = 5(1) + 4(0) = 5$

Ask anything

12:52 Fri, 4 Apr 74%

ChatGPT

- First Equation:
  $x_1 + 2x_2 = 5(1) + 4(0) = 5$.

- Second Equation:
  $x_1 + 3x_2 = 6(1) + 3(0) = 6$.
- Third Equation:
  $x_1 + 2x_2 = 5(1) + (0) = 5$.

Notice the first and third equations are identical, so we really have only two independent equations for the two unknowns x_1 and x_2.

---

4. Solving for $\mathbf{x} = (x_1, x_2)$

We now solve the simplified system:

LaTeX Copy code

```
\begin{cases}
x_1 + 2x_2 = 5,\鉾1mm]
x_1 + 3x_2 = 6.
\end{cases}
```

Subtract the first equation from the second:

$$(x_1 + 3x_2) - (x_1 + 2x_2) = 6 - 5,$$

$$x_2 = 1.$$

Substitute $x_2 = 1$ into the first equation:

$$x_1 + 2(1) = 5 \quad \Longrightarrow \quad x_1 = 5 - 2 = 3.$$

So, we obtain:

$$\mathbf{x} = (3, 1).$$

---

5. Summary

Ask anything

(15) p+q>n 참!

↓

some cols comb^n of others.

Some nonzero vector in V can be written as comb^n of W's basis vectors.

which is in the intersection $V \cap W$

we know, $\dim(V \cap W) \geq p+q-n$.

since, $p+q > n$, $\dim(V \cap W) > 1$

↓

At least 1 nonzero vector

$\rightarrow 1$.

This is an important geometric insight because it tells us that in higher-dimensional spaces, it is likely that two subspaces will intersect rather than being completely disjoint.

then ↓ if ↗

$\{0\}$ intersect at 0 vector.

↓

trivial intersection.

(16) $A^T y = 0$

$Ax = b$

~~$b^T y = 0$~~

$(Ax)^T y = 0$

$x^T A^T y = 0$

$\therefore y \perp Ax$ & $N(A^T) \perp C(A)$

(17) S spanned by $(1,1,1)$ → A line thru the origin in R^3

A vector $v(x,y,z)$ belongs to $S^\perp$ if & only if it is orthogonal to every vector in S. Since every vect in S is scalar multiple of $(1,1,1)$ → (spanned by $1,1,1$),

$$v.(1,1,1) = x+y+z=0$$

↓ eqn of a plane in R^3

$(1,-1,0) = 0$ Soln

$(1,0,-1) = 0$

S spanned by $(1,1,1)$ & $(1,1,-1)$

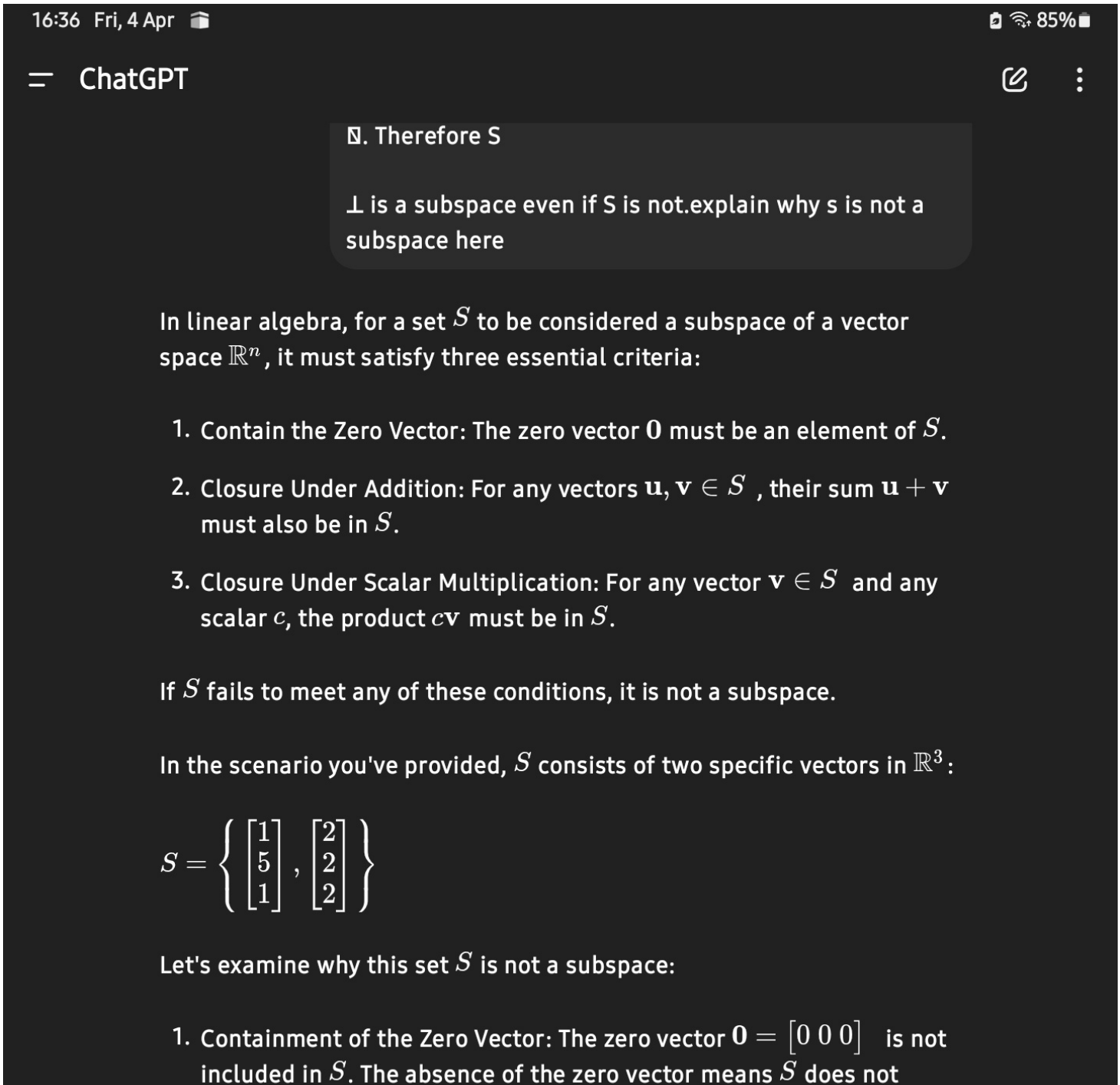
So, $v.(1,1,1) = x+y+z=0$

$x+y=0$

$x=-y$

$\therefore$ Ans. $(1,-1,0)$

(18)

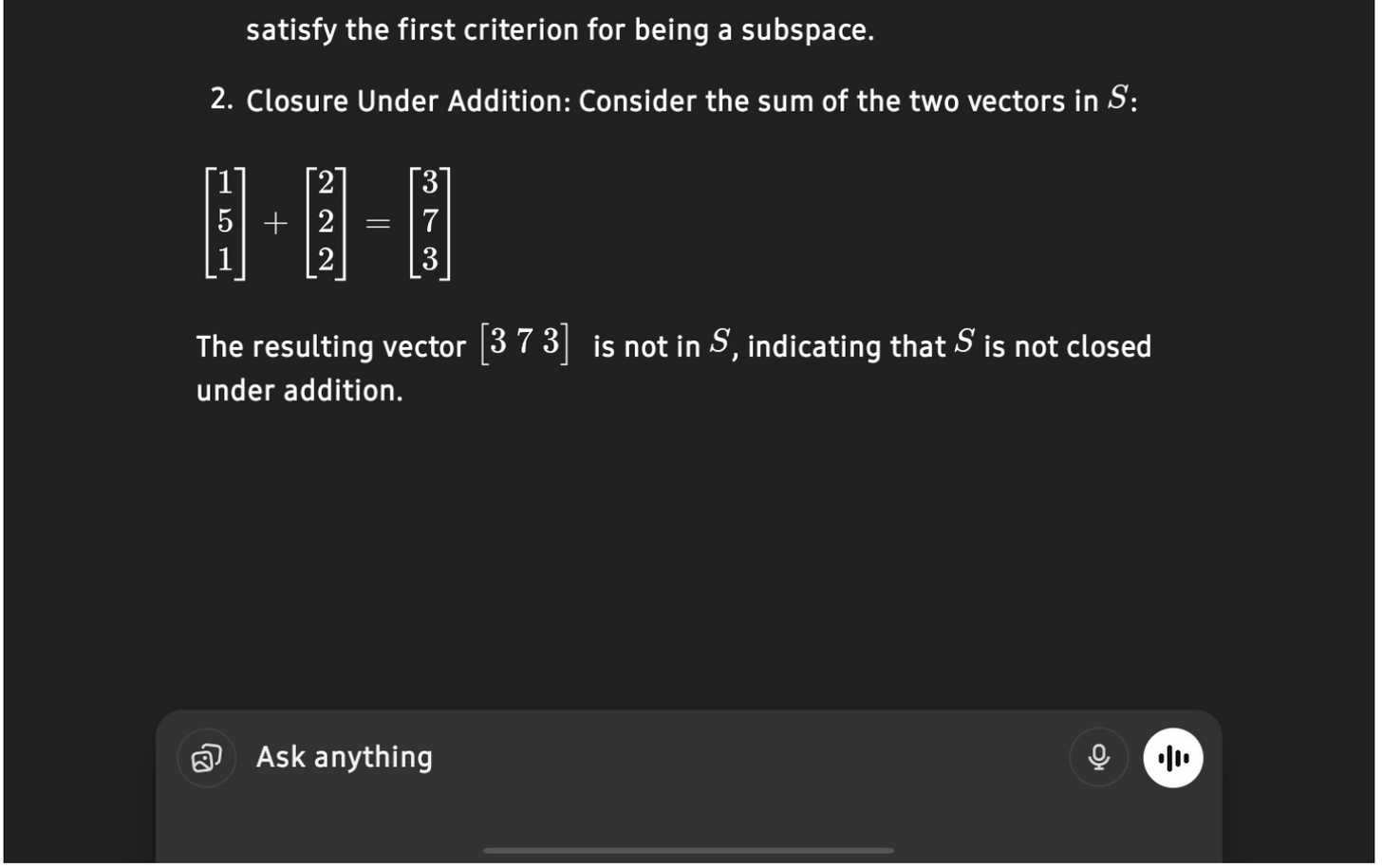


satisfy the first criterion for being a subspace.

2. Closure Under Addition: Consider the sum of the two vectors in S:

$$\begin{bmatrix}1\\5\\1\end{bmatrix} + \begin{bmatrix}2\\2\\2\end{bmatrix} = \begin{bmatrix}3\\7\\3\end{bmatrix}$$

The resulting vector $\begin{bmatrix}3 & 7 & 3\end{bmatrix}$ is not in S, indicating that S is not closed under addition.



In contrast, the null space (or kernel) of a matrix A, defined as xEV Ax-0 , always satisfies all three conditions:

Zero Vector: A0 =0

so 0 is in the null space

Closed Under Addition: If Au= 0 and Av = 0, then A(u+v)=Aut Av=0+0=0 ,soutv isinthe null space.

• Closed Under Scalar Multiplication: If Av = 0 and c is any scalar

then A(cv) =c(Av) = c0 = 0 so cv is in the null space

Therefore, the null space of A is always a subspace of V

(21) S is spanned by $v_1 = (1,2,2,3)$, $v_2 = (1,3,3,2)$

$S^{\perp}$ consists of all vect $\perp$ to all vect to S.

$$v_i \cdot x = 0 \qquad \forall v_i \in S$$

So, $x_1 + 2x_2 + 2x_3 + 3x_4 = 0$

$x_1 + 3x_2 + 3x_3 + 2x_4 = 0$

Now,

$$\begin{bmatrix} 1 & 2 & 2 & 3 \\ 1 & 3 & 3 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$E_{21}A = \begin{bmatrix} 1 & 2 & 2 & 3 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

Now the system:

$x_1 + 2x_2 + 2x_3 + 3x_4 = 0$ — (I)

$x_2 + x_3 - x_4 = 0$ — (II)

$\therefore x_2 = -x_3 + x_4$

(1) $\Rightarrow$

$$x_1 - 2x_3 + 2x_4 + 2x_3 + 3x_4 = 0$$

$$x_1 + 5x_4 = 0 \qquad \therefore x_1 = -5x_4$$

taking $x_3 = 0, \quad x_4 = 1,$

$$x_2 = 1 \quad , \quad x_1 = -5$$

$$\therefore \begin{bmatrix} -5 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

taking $x_3 = 1, \quad x_4 = 0$

$$x_2 = 1 \quad , \quad x_1 = 0$$

$$\therefore \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = s \begin{bmatrix} -5 \\ 1 \\ 0 \\ 1 \end{bmatrix} + t \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}$$

(23)

If a subspace S is contained in a subspace V, explain why S* contains V*.

=>

x in V⊥ is perpendicular to every vector in V. Since V contains all the vectors in S,x is perpendicular to every vector in S. So every x in V⊥ is also in S⊥.

(24) $AA^{-1} = I$. 1st col of A^{-1} $\begin{bmatrix}1\\0\\0\\\vdots\end{bmatrix}$ is ⊥ to 2, 3 . . . . n rows of A.

(25) $A^TA = I$

(26)

$$\begin{bmatrix} 1 & 2 \\ 2 & 2 \\ 3 & -2 \end{bmatrix}_{3\times 3}$$

17:38 Fri, 4 Apr 86%

ChatGPT

$$A^TA = \begin{bmatrix} 2 & 2 & -1 \\ 2 & 2 & -1 \\ -1 & 2 & 2 \end{bmatrix} \times \begin{bmatrix} 2 & 2 & -1 \\ -1 & 2 & 2 \\ 2 & -1 & 2 \end{bmatrix}$$

$$A^TA = 9I = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

- If the columns were unit vectors, then $A^TA = I$, meaning the

matrix is orthogonal.

Explanation of Question 27:

You need to analyze the lines given by:

$3x + y = b_1$

6x + 2y = b_2]

- Parallel lines condition:
  - The second equation can be rewritten as $6x + 2y = b_2$, which is just double the first equation.
  - This means the two lines are parallel because their normal vectors $(3, 1)$ and $(6, 2)$ are proportional.
- When are the lines identical?
  - If $b_2 = 2b_1$, the second equation is just a scaled version of the first, meaning they represent the same line.
- Nullspace of the matrix:
  - The system can be rewritten in matrix form as:

$$\begin{vmatrix} 3 & 1 \\ 6 & 2 \end{vmatrix} \begin{vmatrix} x \\ y \end{vmatrix} = \begin{vmatrix} b_1 \\ b_2 \end{vmatrix}$$

$$\begin{vmatrix} 3 & 1 \\ 6 & 2 \end{vmatrix} \begin{vmatrix} x \\ y \end{vmatrix} = 0$$

- Nullspace interpretation:
  - Any vector in the nullspace satisfies $3x + y = 0$, meaning the solution set is a line in 2D.
  - A particular vector in the nullspace is $(-1, 3)$, meaning moving along this direction preserves solutions in the nullspace.

Ask anything

Problem 26:
To construct a 3x3 matrix A with mutually perpendicular columns and no zero entries:

1. Choose orthogonal columns:

- Column 1: $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$\begin{bmatrix} 3 \end{bmatrix}$

- Column 2: $\begin{bmatrix} 3 \\ 3 \\ -3 \end{bmatrix}$ (orthogonal to column 1)
- Column 3: $\begin{bmatrix} 5 \\ -4 \\ 1 \end{bmatrix}$ (orthogonal to columns 1 and 2).

2. Matrix A:

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 3 & -4 \\ 3 & -3 & 1 \end{bmatrix}$$

3. Compute $A^T A$:
   - Columns are orthogonal, so $A^T A$ is diagonal with entries equal to the squared norms of the columns:

$$A^T A = \begin{bmatrix} 14 & 0 & 0 \\ 0 & 27 & 0 \\ 0 & 0 & 42 \end{bmatrix}$$

   Explanation: The off-diagonal entries are zero because the columns are mutually perpendicular (dot product = 0).

Problem 27:
The lines $3x + y = b_1$ and $6x + 2y = b_2$:

1. Parallel Lines: The second equation is a scalar multiple of the first (\(6x + 2y = 2(3x + y) \), so they are **parallel**.
2. Same Line Condition: They coincide if $b_2 = 2b_1$.
3. Perpendicular Vector: When $b_2 = 2b_1$, the vector (b_1, b_2) is proportional to $(1, 2)$, which is perpendicular to $(-2, 1)$.
4. Nullspace: The nullspace of the coefficient matrix $\begin{bmatrix} 3 & 1 \\ 6 & 2 \end{bmatrix}$ is the line $3x + y = 0$.
5. Example Vector in Nullspace: $(-1, 3)$ satisfies $3(-1) + 1(3) = 0$.

Final Answers

New chat

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DeepThink (R1) Search

17:39 Fri, 4 Apr 86%

Nullspace and Subspace Analysis Explained

First, the lines are $3x + y = b_1$ and $6x + 2y = b_2$. When they are the same line, the second equation is just a multiple of the first. So, $6x + 2y = b_2$ must be $2*(3x + y) = 2b_1$, hence $b_2 = 2b_1$. That makes sense.

Now, the question says that in this case, the vector (b_1, b_2) is perpendicular to (-2, 1). Why? Let's think about the relationship between the coefficients of the equations and

the vectors.

The normal vector to the line $3x + y = b_1$ is (3, 1), right? Because the coefficients of x and y in the equation ax + by = c give the normal vector (a, b). Similarly, for $6x + 2y = b_2$, the normal vector is (6, 2), which is just 2*(3, 1). So, the normal vectors are parallel, confirming the lines are parallel or coinciding.

But why is (b_1, b_2) perpendicular to (-2, 1)? Maybe it's related to the condition for consistency of the system. If the lines are the same, the equations must be consistent. For the system to be consistent, the vector (b_1, b_2) must lie in the column space of the coefficient matrix. Wait, the coefficient matrix here is:

[[3, 1],
[6, 2]]

Which has rank 1 because the rows are multiples of each other. So, the column space is spanned by (3, 6) or (1, 2). The vector (b_1, b_2) must be a multiple of (1, 2) to be in the column space. So, if $b_2 = 2b_1$, then $(b_1, b_2) = b_1$*(1, 2).

Now, the vector (1, 2) is perpendicular to (-2, 1) because their dot product is (1)(-2) + (2)(1) = -2 + 2 = 0. Ah! So, if (b_1, b_2) is a multiple of (1, 2), then it's automatically perpendicular to (-2, 1). Therefore, when the lines coincide, (b_1, b_2) is parallel to (1, 2), which is orthogonal to (-2, 1).

Wait, but the problem states that (b_1, b_2) is perpendicular to (-2, 1). If (b_1, b_2) is a scalar multiple of (1, 2), then yes, because (1, 2) · (-2, 1) = 0. So, any scalar multiple of (1, 2) would also be perpendicular to (-2, 1). Therefore, when $b_2 = 2b_1$, the vector (b_1, b_2) is in the direction of (1, 2), making it perpendicular to (-2, 1). That makes sense now.

When the lines $3x + y = b_1$ and $6x + 2y = b_2$ coincide (b_2 = 2b_1\)), the vector \((b_1, b_2)\) becomes a scalar multiple of \(1, 2)\). The vector \((1, 2)\) is orthogonal to \((-2, 1)\) because their dot product is:

$$(1)(-2) + (2)(1) = -2 + 2 = 0.$$

Thus, \(b_1, b_2)\) lies in the direction of \((1, 2)\), ensuring it is perpendicular to \((-2, 1)\).

Final Answer

$\boxed{(b_1, b_2)}$ is perpendicular to $\boxed{(-2, 1)}$ because when $b_2 = 2b_1$, \(b_1, b_2)\) aligns with \((1, 2)\), which is orthogonal to \((-2, 1)\).

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DeepThink (R1) Search

32) $r^T n = 0$, $c^T l = 0$, $A = acr^T$, $a \neq 0$

$[c_1 c_2]^T M [r_1 r_2]^T$ → 2x2 invertible m

33